

Difference Between Schrödinger Equation and Master Equation

May 6, 2026

1 Introduction

In quantum mechanics, the evolution of systems can be described using either the **Schrödinger equation** (for closed systems) or **master equations** (for open systems). Understanding the distinction is crucial for modern physics, especially quantum computing, quantum optics, and statistical mechanics.

2 Schrödinger Equation

2.1 Wavefunction Form

$$i\hbar \frac{\partial}{\partial t} \psi(t) = H\psi(t) \quad (1)$$

2.2 Density Matrix Form (von Neumann Equation)

$$\frac{d\rho}{dt} = -\frac{i}{\hbar} [H, \rho] \quad (2)$$

2.3 Key Properties

- Describes **closed systems**
- Evolution is **unitary**
- Pure states remain pure
- No decoherence or dissipation
- Time-reversible dynamics

3 Master Equation

For open quantum systems interacting with an environment, the evolution is given by the Lindblad master equation:

$$\frac{d\rho}{dt} = -\frac{i}{\hbar} [H, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right) \quad (3)$$

3.1 Key Properties

- Describes **open systems**
- Non-unitary evolution
- Pure states can become mixed
- Includes decoherence and dissipation
- Irreversible dynamics

4 Physical Meaning of Terms

- H : System Hamiltonian
- L_k : Lindblad (jump) operators representing environment interactions
- $\{A, B\} = AB + BA$: Anticommutator

5 Example: Two-Level System (Qubit)

5.1 Hamiltonian

$$H = \frac{\hbar\omega}{2}\sigma_z \quad (4)$$

5.2 Initial State

$$\rho(0) = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \quad (5)$$

6 Case A: Schrödinger Evolution

$$\rho(t) = e^{-iHt/\hbar}\rho(0)e^{iHt/\hbar} \quad (6)$$

$$\rho(t) = \begin{pmatrix} \frac{1}{2} & \frac{1}{2}e^{-i\omega t} \\ \frac{1}{2}e^{i\omega t} & \frac{1}{2} \end{pmatrix} \quad (7)$$

Insight: Coherence (off-diagonal terms) oscillates but does not decay.

7 Case B: Master Equation (Dephasing)

Let the Lindblad operator be:

$$L = \sqrt{\gamma}\sigma_z \quad (8)$$

Then the master equation becomes:

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H, \rho] + \gamma(\sigma_z\rho\sigma_z - \rho) \quad (9)$$

7.1 Solution

$$\rho(t) = \begin{pmatrix} \frac{1}{2} & \frac{1}{2}e^{-i\omega t}e^{-2\gamma t} \\ \frac{1}{2}e^{i\omega t}e^{-2\gamma t} & \frac{1}{2} \end{pmatrix} \quad (10)$$

Insight: Coherence decays exponentially due to environment interaction.

8 Numerical Illustration

Let:

$$\omega = 1, \quad \gamma = 0.2, \quad t = 5$$

8.1 Schrödinger Evolution

$$|\rho_{12}| = 0.5 \quad (11)$$

8.2 Master Equation

$$|\rho_{12}| = 0.5e^{-2\gamma t} = 0.5e^{-2} \approx 0.068 \quad (12)$$

Interpretation: Coherence reduces drastically, indicating decoherence.

9 Conceptual Insight

9.1 Why Master Equation is Needed

Real systems interact with environments. The total system evolves as:

$$\rho_{\text{system}} = \text{Tr}_{\text{env}}(\rho_{\text{total}}) \quad (13)$$

Tracing out the environment leads to:

- Loss of coherence
- Irreversibility
- Mixed states

10 Classical Analogy

The classical master equation:

$$\frac{dP_i}{dt} = \sum_j (W_{ij}P_j - W_{ji}P_i) \quad (14)$$

The Lindblad equation is its quantum generalization.

11 Comparison Summary

Feature	Schrödinger	Master Equation
System Type	Closed	Open
Evolution	Unitary	Non-unitary
Decoherence	No	Yes
Reversibility	Yes	No
States	Pure only	Mixed allowed

12 Q&A

Q1. Why can't Schrödinger equation describe decoherence?

Because it preserves purity:

$$\rho(t) = U\rho(0)U^\dagger$$

No entropy increase is possible.

Q2. What do Lindblad operators represent?

They represent environmental interactions such as:

- Noise
- Measurement
- Dissipation

Q3. Can master equations be derived from Schrödinger equation?

Yes:

1. Start with system + environment
2. Apply Schrödinger equation
3. Trace out environment
4. Apply approximations (Born-Markov)

Q4. What is lost in master equation?

- Phase information
- System-environment correlations
- Reversibility

Q5. When does master equation fail?

- Strong system-environment coupling
- Non-Markovian dynamics (memory effects)

13 Conclusion

The Schrödinger equation describes ideal, isolated quantum systems, while master equations describe realistic systems interacting with environments. The transition between them explains how classical behavior emerges from quantum mechanics.